DUAL-MCU ARCHITECTURE & FOTA UPDATE SYSTEM

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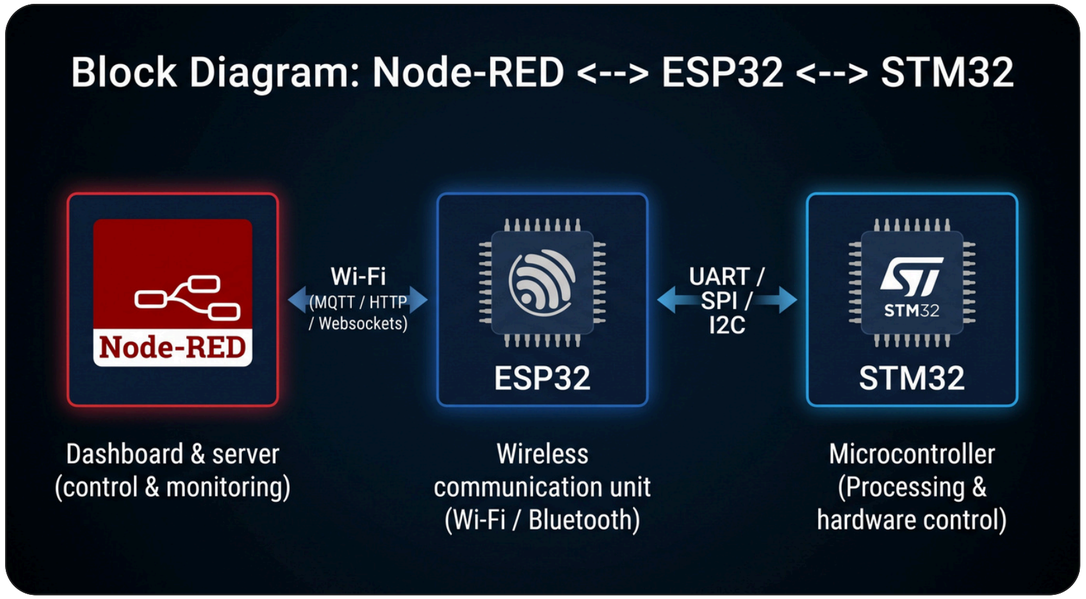
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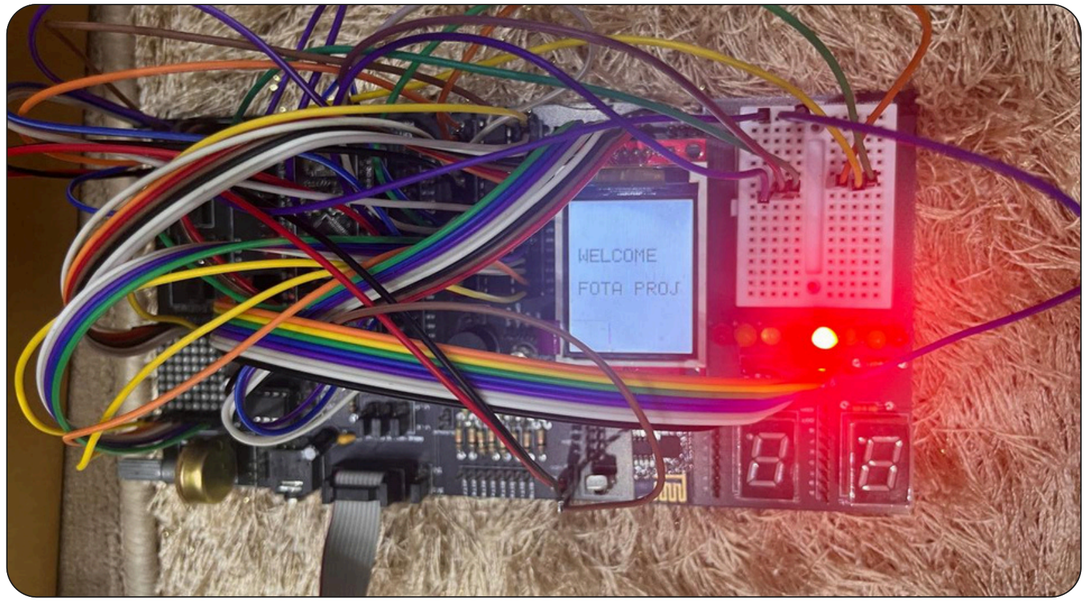
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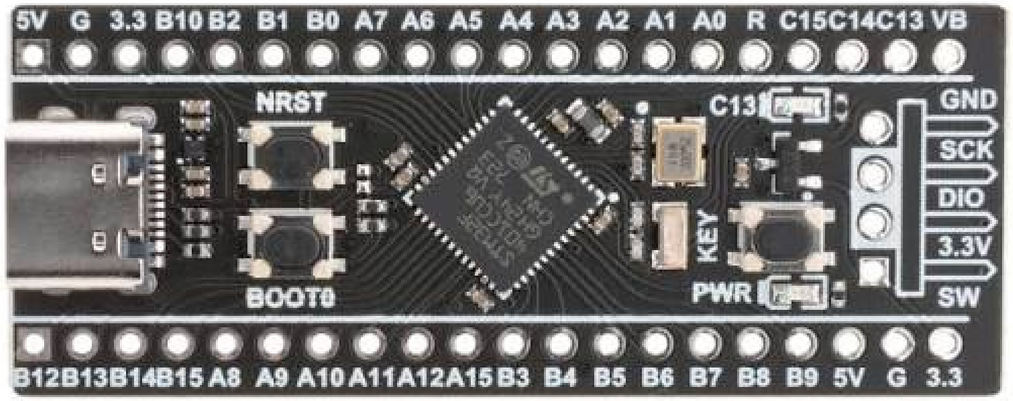
INTRODUCTION





- Wireless Firmware Update (FOTA): Stream and flash new Intel HEX firmware over Wi-Fi to STM32 without physical programmers.
- Dual-Control System: Real-time local control via NEC IR Remote (EXTI + SysTick) and remote control via Node-RED Dashboard.
- Synchronized Visual : 5 dynamic animation modes (Chaser, Ping-Pong, Sorting, Bubbles, Blink) across 8 LEDs, an 8x8 LED Matrix (via 74HC595 STP), and a 1.8" TFT Display.

HARDWARE DESIGN & MCUS





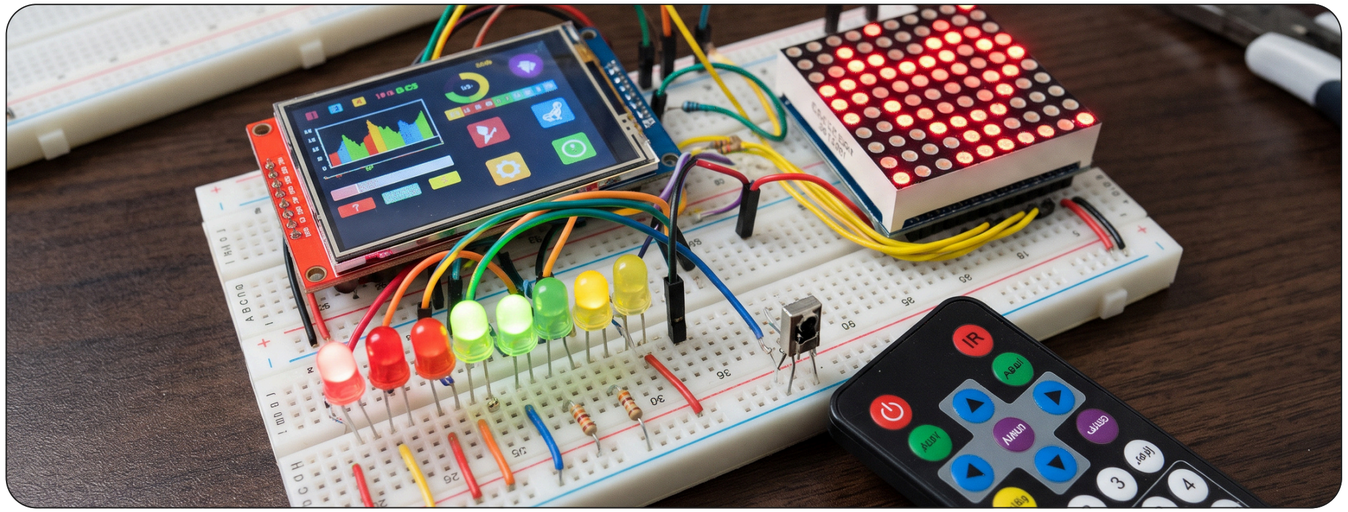
1. STM32F401CC (ARM Cortex-M4): Runs the custom Bootloader (Sectors 0–1, 32 KB) and Main Application (Sectors 2–5, 224 KB), handling real-time display rendering, IR decoding, and LED animations.

2. ESP32 Wi-Fi Module: Acts as the IoT Gateway and HTTP Web Server (Port 80), storing the firmware HEX array in flash and managing the non-blocking FOTA state machine.

UART COMMUNICATION LINK

Full-duplex serial link (9600 bps) connecting STM32 USART1 (PA9/PA10) with ESP32 UART2 (GPIO 16/17) forHEX streaming and real-time commands.

PERIPHERALS & INTERFACING



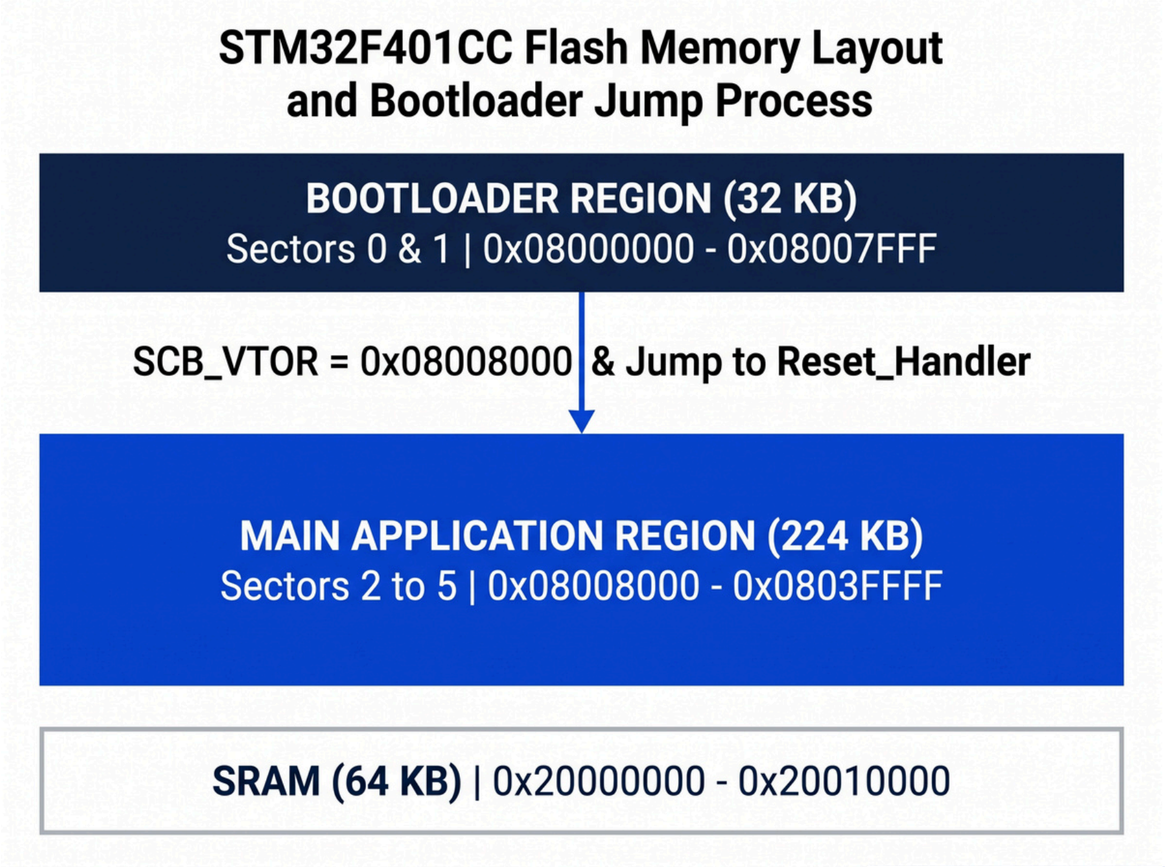
Input =>
NEC IR Receiver (PA1) Captures remote control signals via External Interrupt (EXTI1) for local mode switching.
Active Buzzer (PA8) Provides non-blocking audible confirmation tones on every command.

Visual Output Units =>
1.8" TFT Screen Driven via SPI1 (PA5, PA7) + Control Pins (PA0, PA2, PA4) for live mode & FOTA status.
8x8 LED Matrix & 8 LEDs Matrix driven via 74HC595 Shift Register (PB12, PB13, PB15) alongside 8 status/animation LEDs (PB0–PB7).

SOFTWARE LAYERED ARCHITECTURE & MEMORY MAP

STM32F401CC Flash is partitioned into 32 KB for the Custom Bootloader (Sectors 0–1) and 224 KB for the Main Application (Sectors 2–5).

- APP Layer: FOTA State Machine, IR Remote Handler, LED Animations, Welcome & Mode Displays.
- HAL Layer: 1.8" TFT Screen (RGB565), 8x8 LED Matrix (74HC595 STP), Dynamic 8-LED Array, Active Buzzer.
- MCAL Layer: Custom register-level drivers for RCC, GPIO, EXTI1 + NVIC + SysTick (pulse-width timing), SPI1 + DMA2, USART1, and FMI (Flash Memory Interface).
- ESP32 Gateway Software: HTTP WebServer running a non-blocking FOTA State Machine (IDLE, ERASING, SENDING, DONE).

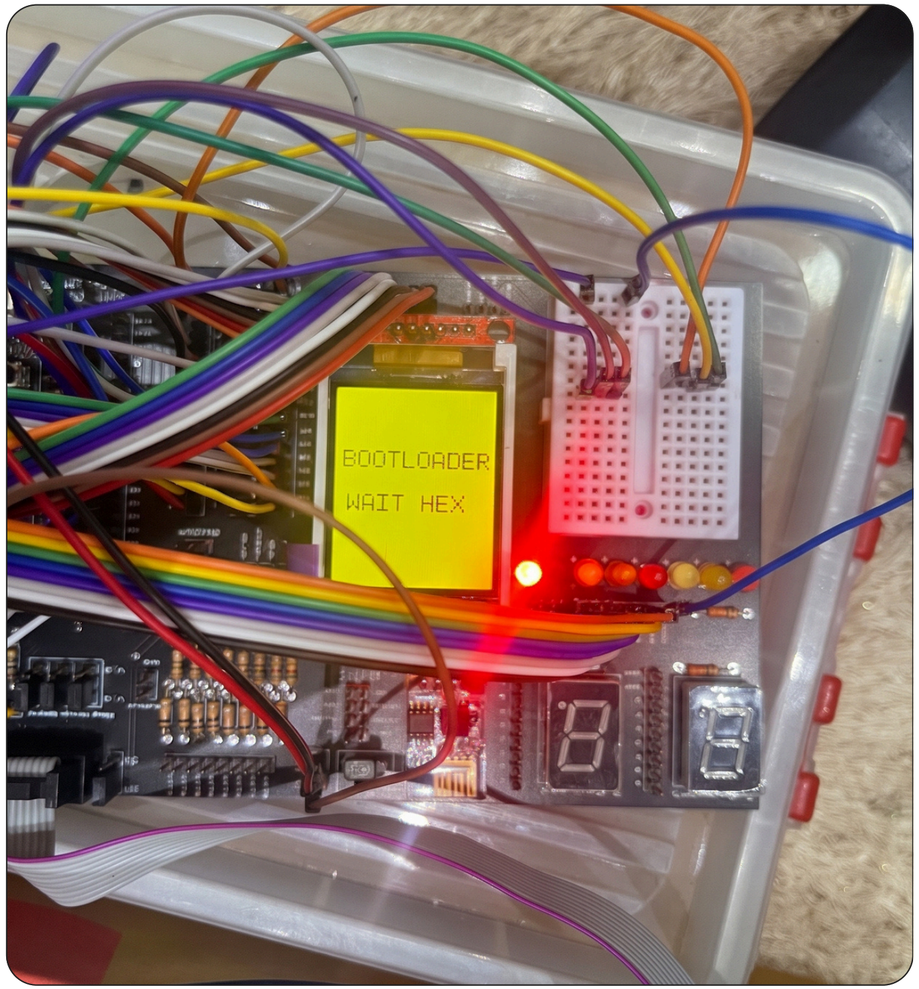


SYSTEM OPERATION & SEQUENCES

- 1. Erase Phase ('E'): User clicks Start FOTA on Node-RED; ESP32 sends 'E' over UART; STM32 Bootloader erases Sectors 2–5 via MFMI and replies "OK".

- 2. HEX Streaming (':'): ESP32 streams Intel HEX lines one by one; STM32 parses and flashes each record with "OK" handshaking and live progress polling (0% to 100%).

- 3. EOF & Handover: Upon receiving :00000001FF, Bootloader validates MSP, sets SCB_VTOR = 0x08008000, and jumps to the Main App.



SYSTEM OPERATION & SEQUENCES

- 1. Initialization: Once jumping to 0x08008000, the Main App initializes RCC, GPIO, SPI1 (TFT), STP (8x8 Matrix), and EXTI1 + SysTick.

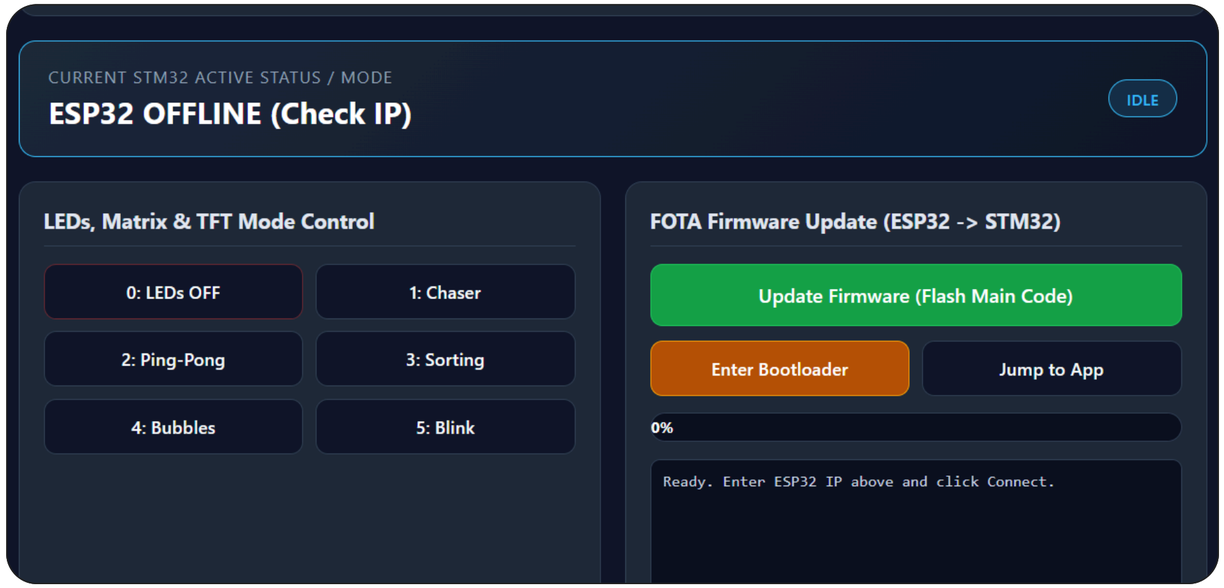
- 2. IR & Command Capture: Falling edges on PA1 trigger EXTI1 to measure pulse widths via SysTick, decoding the 8-bit NEC command (Bits 17–24) alongside UART polling, triggering a Buzzer beep on PA8.

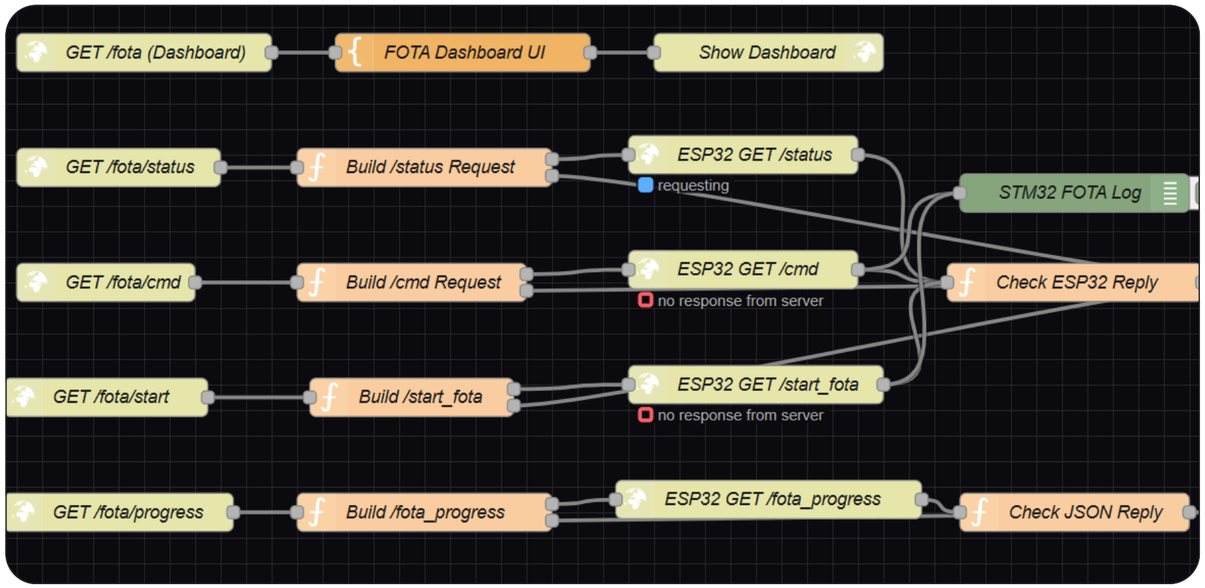
- 3. Visual Sync & Non-Blocking Modes: Updates TFT (RGB565) & 8x8 Matrix while running 5 non-blocking LED modes (Chaser | Ping-Pong | Sorting | Bubbles | Blink).

- 4. Live Status Feedback: Continuous UART status synchronization with ESP32 & Node-RED.



IOT DASHBOARD & PROJECT MILESTONES





- Mode Selection Control: Instant buttons to switch between the 5 LED/Matrix/TFT modes or turn outputs OFF remotely.
- FOTA Firmware Upgrade Panel: Dedicated controls to trigger Firmware Update, Enter Bootloader, or Jump to App with a real-time percentage progress bar.

IOT DASHBOARD & PROJECT MILESTONES

1. INITIALIZATION >>>

Bootloader validates application and jumps to memory, initializing standard interfaces (SPI, UART, EXTI) and showing the startup greeting.

2. REMOTE OPERATION >>>

The system captures local IR commands or processes HTTP gateway requests, seamlessly updating modes, TFT displays, and visual matrixes.

3. FOTA FLASHER >>>

Triggering updates through Node-RED launches Sector erase & line-by-line file upload, showcasing live status monitoring before self-boot.

THANK YOU!

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